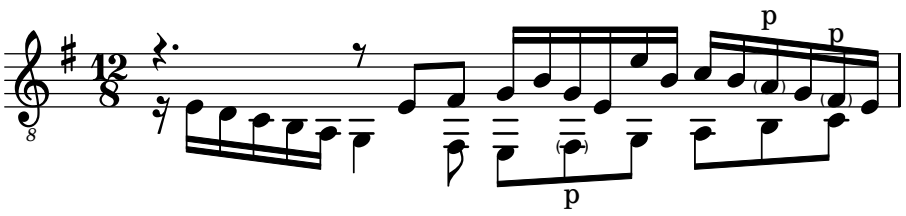
GRATIS AD PARNASSUM

A Free Workbook for 18th-Century Counterpoint



by Jonathan Kulp

GRATIS AD PARNASSUM

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Preface

I decided to write this workbook in response to the escalating price of textbooks. I found that my students were paying upwards of $60 for a workbook with no more than fifty pages in it, and felt that I could probably write one just as good myself and give it to them for free. For several years I have allowed my students to purchase old editions of Kennan's Counterpoint textbook,[1] and have been pleased that they can normally find copies online for under $10. Other than pagination, the old editions are practically identical to the most recent edition, which now costs more than $120 at retail price. This workbook is not intended to be a textbook. I include some text and tips and guidance, but I still require students to purchase the Kennan textbook.

My decision to give the book away for free is the result of two influences in my life. First of all, the fact that I was recently granted tenure in my job at the University of Louisiana at Lafayette means that the pressure to publish in the traditional sense is less severe. Secondly, I have become fairly involved in the free and open-source software (a.k.a. "FOSS") movement, represented most visibly by the various flavors of the **GNU/Linux** operating system, but also by smaller projects and developers around the world who are devoted to producing excellent software and sharing it freely with others.

One of the primary tenets of open-source development is that individual users may see the source code and are free to make improvements upon it, but those improvements must be shared back to the community. In that spirit I welcome contributions to this project in the form of corrections, additions or other help. I encourage you to use the workbook as you see fit and to share it freely with anyone who wants it. In return I ask only that you contact me and let me know if you're using it so that I can acknowledge this in the book and so that hopefully my institution's administrators will see my work's value to the academic community.

I would like to thank the students in my Spring 2009 Counterpoint class for being guinea pigs in using the first version of this workbook: Nathan Abbott, Andy Angelle, Seth Bagwell, Danny Devillier, Beverly Harlton, Brandon LeJeune, Tom Lyles, Nicholas Manzo, Sasha Massey, and Leigh Stewart. Thanks y'all!

—JLK

[1] Kent Kennan, *Counterpoint*, 4th ed. (Englewood Cliffs, NJ: Prentice-Hall, 1999).

Chapter 1
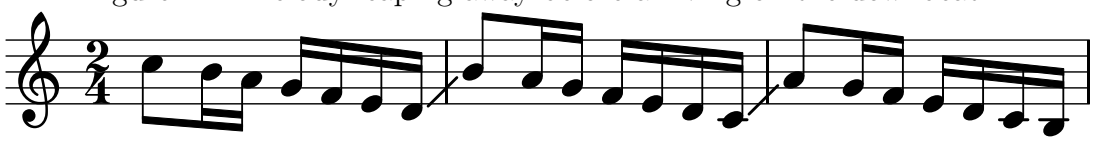
Melody

Since counterpoint is by definition the combination of melodies, it goes without saying that counterpoint is better when the melodies that comprise it are themselves good. According to conventions of 18th–century counterpoint, good melodies have the following characteristics:

- Pleasing contour
  - Clear sense of direction
  - Climax point
  - More steps than leaps
- Logical harmonic implications
  - Tendency tones generally need to follow their tendencies: $\hat{7}$–$\hat{1}$, $\hat{4}$–$\hat{3}$[1]
  - Minor modes: 1) Don't forget leading tone; 2) Use melodic and harmonic minor scales as necessary
- Rhythmic continuity

The preceding is a summary of what Percy Goetschius says about melody in *Elementary Counterpoint*. Please read this section of his book carefully! In addition, I would make the following observation about melodic leaps. Students frequently write melodies that leap in the opposite direction just as the melody is about to arrive on a downbeat.
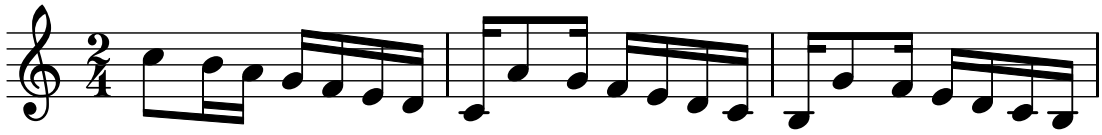
Figure 1.1: Melody leaping away before arriving on the downbeat.



[1]Goetschius, *Applied Counterpoint*, paragraph 6; *Elementary Counterpoint*, par. 6, rule 1.
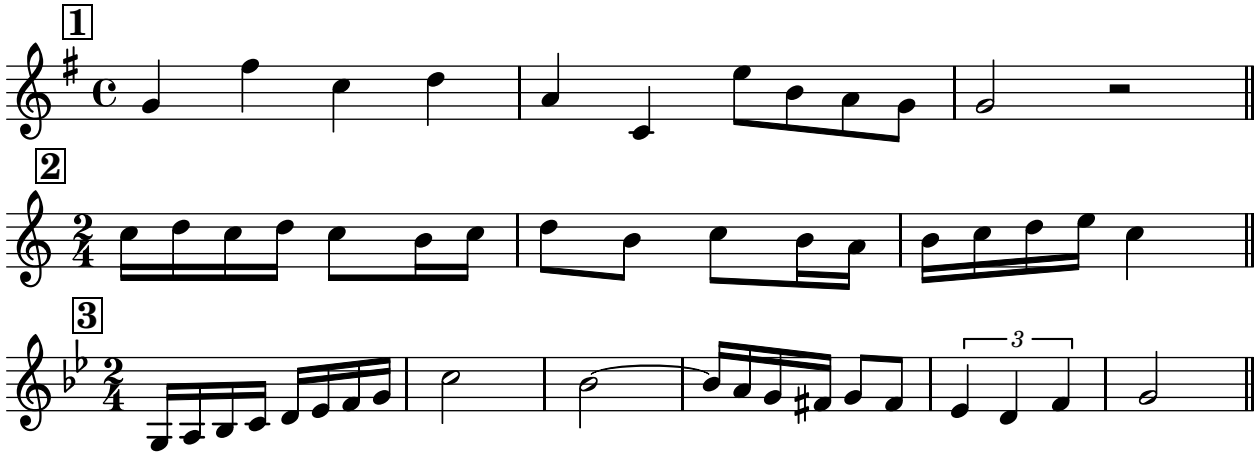
While this is technically not wrong, I find that it is more satisfying in most cases to let the melody continue its descent and land naturally on the downbeat of the next measure before leaping away in the opposite direction. Bach's melodies frequently do this and the results are beautiful. I feel strongly that most of the time the melodies really *want* to land on the downbeat instead of leaping away just before reaching it.[2]

Figure 1.2: Melody lands naturally on downbeat and then leaps in opposite direction.



What's Wrong with This?

Try to find all of the problems in the following examples.



[2]see also Goetschius, *Applied Counterpoint*, paragraph 3*d*, example 8.

Chapter 2

Two-Voice Counterpoint

When writing two-voice counterpoint, several basic principles apply.

- Both melodies should be good by themselves
- Independence
  - Contrasting contours
  - Peaks and valleys happen at different times
  - Contrary Motion
- Coherence
  - Enough in common that they belong together
  - Good harmonic implications
- Rhythmic activity distributed evenly between the voices

Chapter 3

1:1 Counterpoint

First–species counterpoint has one note in the added voice for each note in the cantus firmus. It is in many ways the most challenging kind of counterpoint to write well. There are a number of things to keep in mind when writing first species counterpoint.

- Each voice should be a good melody in itself.[1]
- Imperfect consonances should prevail, perfect intervals should be used with greatest care, normally just as the first and last sonorities in each exercise, or in passing on weak beats.[2]
- While 3^{rds} and 6^{ths} are preferred, try to avoid having more than three in a row of either interval.[3]
- Octaves and unisons should be avoided, since they disrupt the feeling of two voices.[4]
- Repeated notes are not allowed, as they essentially change the relationship of the voices from 1:1 to 2:1.[5]
- Dissonance should be avoided except in rare circumstances.[6]

Tips!

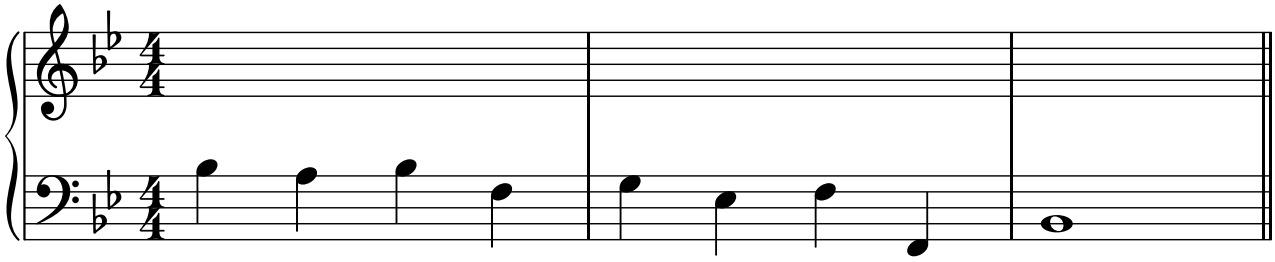
- Play or sing through each cantus firmus three or four times before you try to write any counterpoint against it. This will help form an idea as to what its harmonic implications are.

[1] Goetschius, *Elementary Counterpoint*, paragraph 44, rule 3.
[2] ibid., paragraphs 36, 47.
[3] ibid., paragraph 43, rule 2.
[4] ibid., paragraph 41.
[5] Kennan, *Counterpoint*, p. 37.
[6] For example, I sometimes find it nice to have a 7th at a cadence when the bass has a $\hat{5}$ to $\hat{1}$ motion and the upper voice moves from $\hat{4}$ to $\hat{3}$.

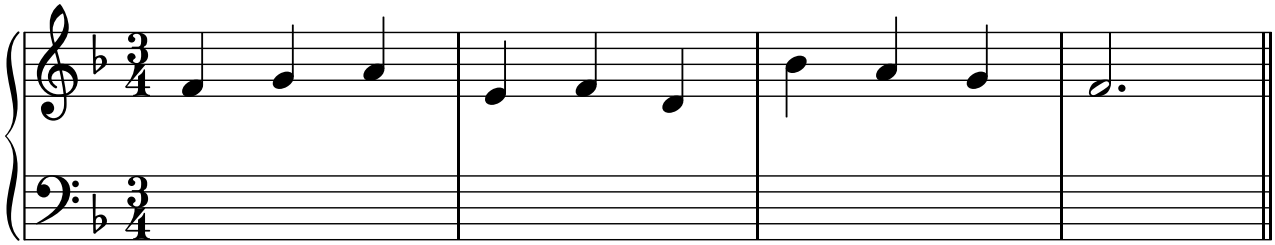
- Play through your completed exercises slowly and carefully on the piano to make sure you have not used perfect intervals inappropriately. I say "slowly" because many students' ears are not yet sensitized to the sound of perfect intervals and easily miss them when played quickly.
- If you find yourself stuck and can't seem to go forward, try starting at the end and working backwards. Make a good cadence and work backwards from there. Sometimes this is very helpful.

Exercises in 1:1 Counterpoint
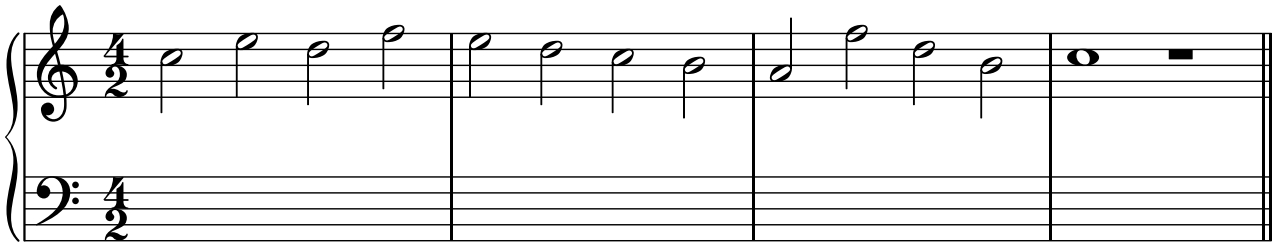
Exercise 1
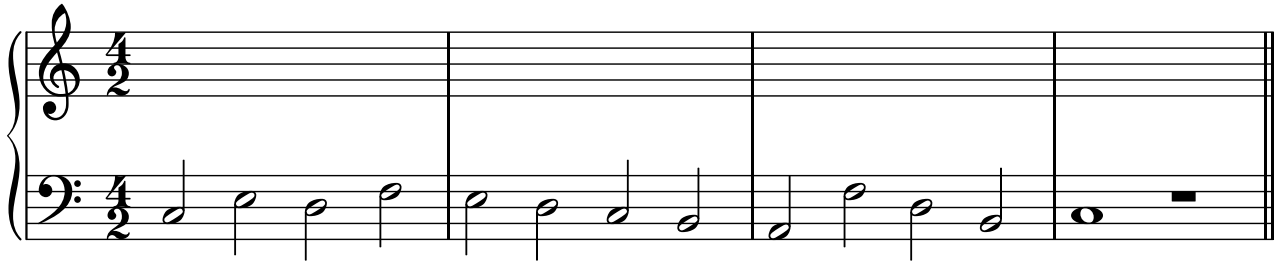


Exercise 2



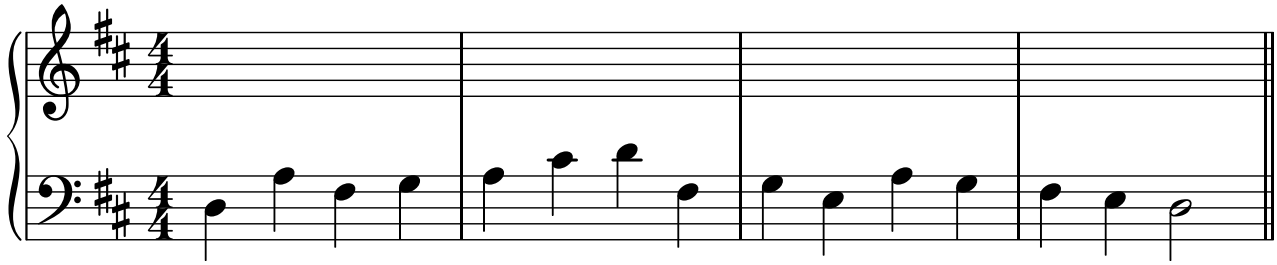
Exercise 3a



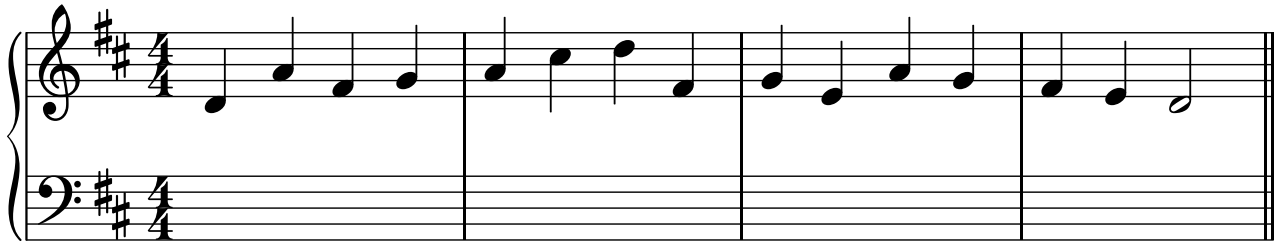
Exercise 3b



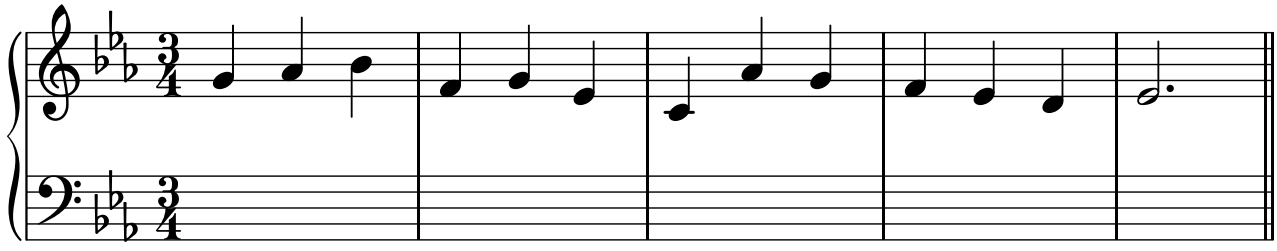
Exercise 4a



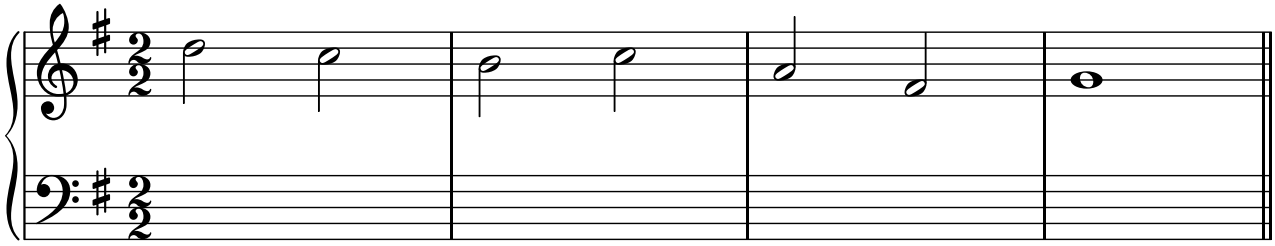
Exercise 4b



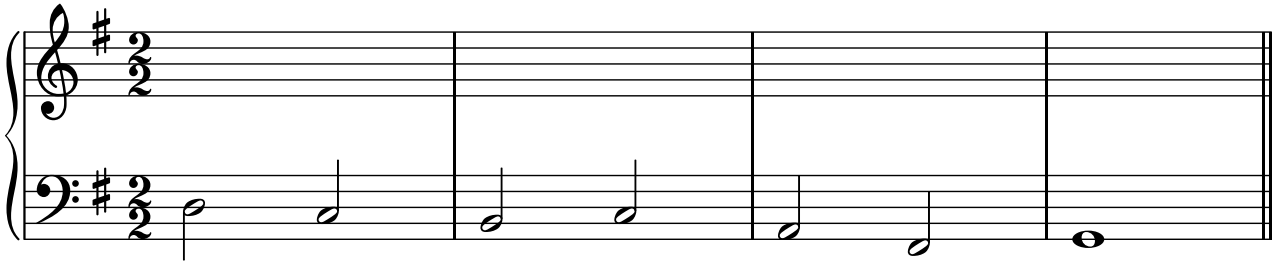
Exercise 5



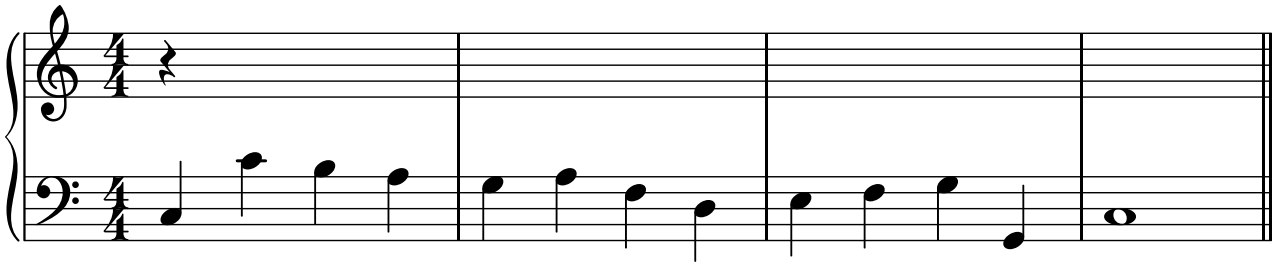
Exercise 6a



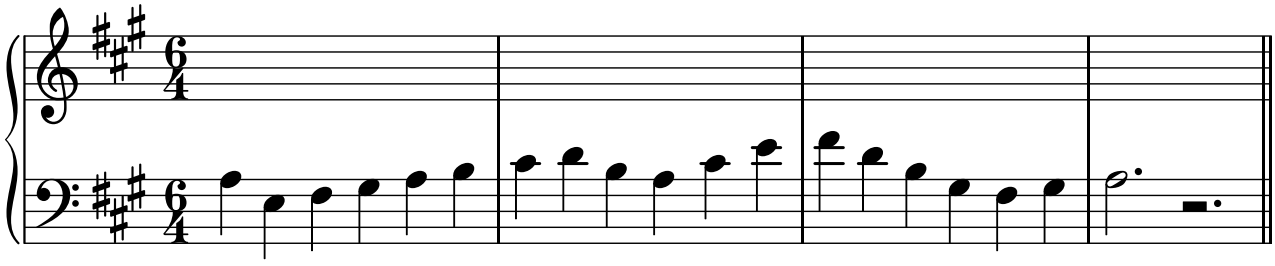
Exercise 6b



Exercise 7



Exercise 8

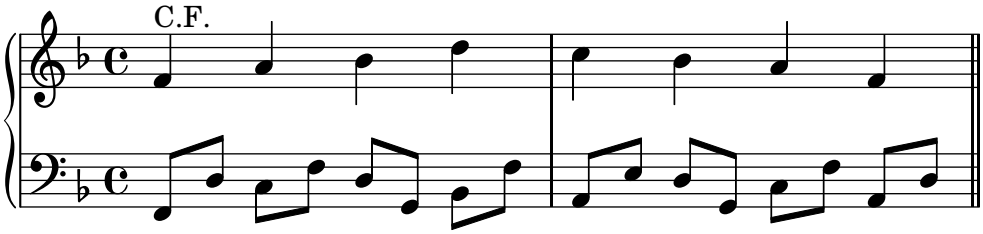


Chapter 4

2:1 Counterpoint

In second species counterpoint there are two notes in one voice for every note in the other. This is one of the most common types of counterpoint. It offers much greater flexibility than first-species counterpoint, since many types of non-chord tones are possible. The availability of non-chord tones makes it much easier to write melodies with smooth contours as well. As such, one should take advantage of this and avoid overly leapy melodies that studiously avoid dissonance. Enjoy your newfound freedom to create dissonance! Here is an example of a student's way-too-leapy exercise.

Figure 4.1: Avoiding dissonance causes leapiness.



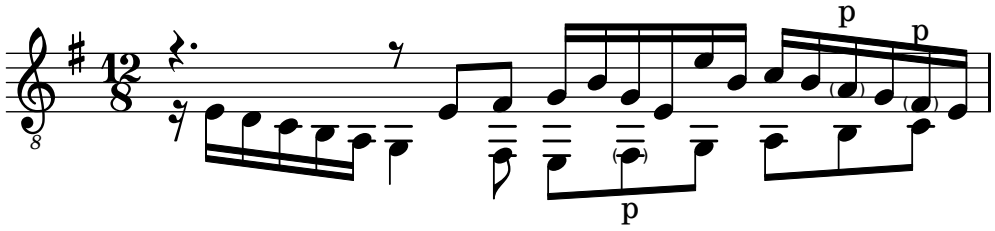
To avoid leaping all over the place, use approved non-chord tones. These will help to smooth out your melodies. For detailed explanations and examples of non-chord tones, see your textbook, but here is a list of common non-chord tones.

- Passing Tones
- Neighbor Tones
- Appoggiaturas
- Escape Tones
- Suspensions (discussed in 4th–species counterpoint)

- Anticipations

Here's a great example of Bach using my favorite non–chord tone, the **accented passing tone**. Accented passing tones happen right on the beat, approached and resolved by step. I strongly encourage their use!
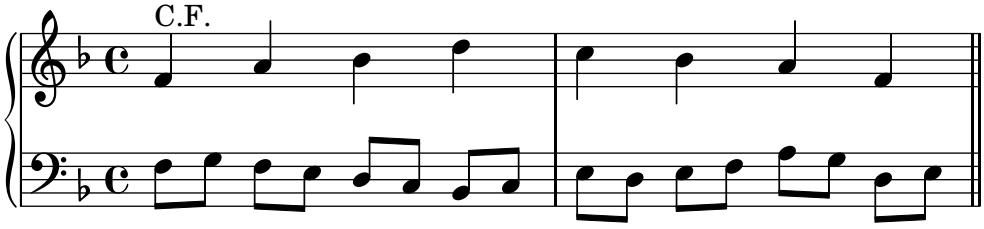
Figure 4.2: Bach: Lute Suite no. 1 in E minor, *Gigue*, m. 1



What's Wrong with This?

Try to determine what's wrong in this example. This is a student's homework in which there are numerous failed attempts at using non–chord tones.

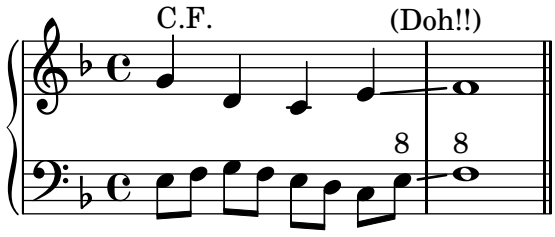
Figure 4.3: Faulty Non–chord Tones



Sneaky Parallels

The problem that afflicts more students than any other in 2:1 counterpoint is what I like to call "sneaky parallels." Sneaky parallels happen when the counterpoint is good **on** the beats, but then on the offbeat one of the voices makes a perfect interval with the other voice, and then moves in parallel to the next beat, creating parallel perfect intervals. This happens almost anywhere but most often at cadences. No matter how many times I warn students about this, they always turn in plenty of sneaky parallels. Here is an example.

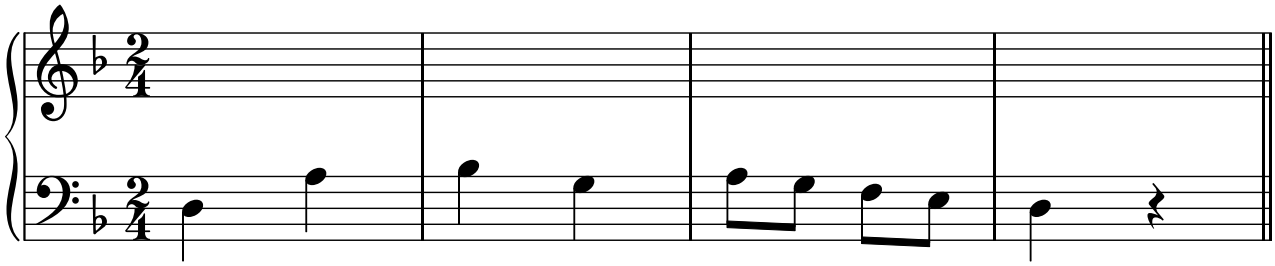
Figure 4.4: Sneaky parallel octaves



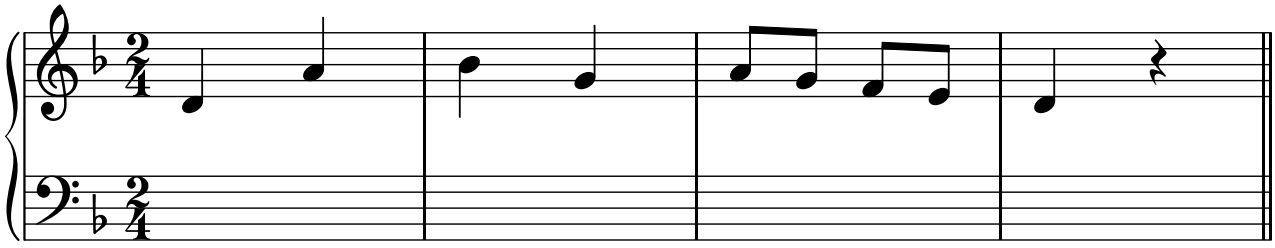
Exercises in 2:1 Counterpoint

Unless otherwise indicated, write two notes in the added voice for each note in the cantus firmus.

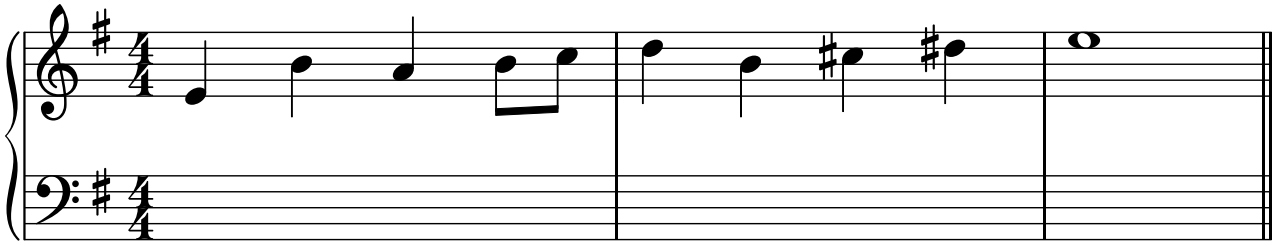
Exercise 1a: When there are 8th notes in the cantus firmus, write quarter or 8th notes in the added voice.



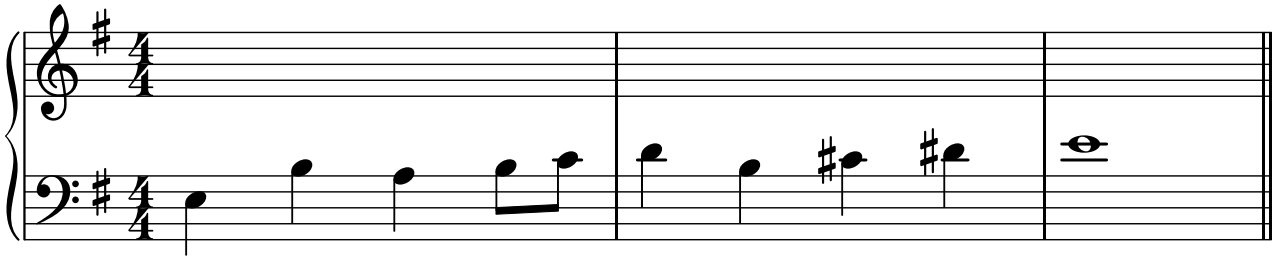
Exercise 1b: When there are 8th notes in the cantus firmus, write quarter or 8th notes in the added voice.
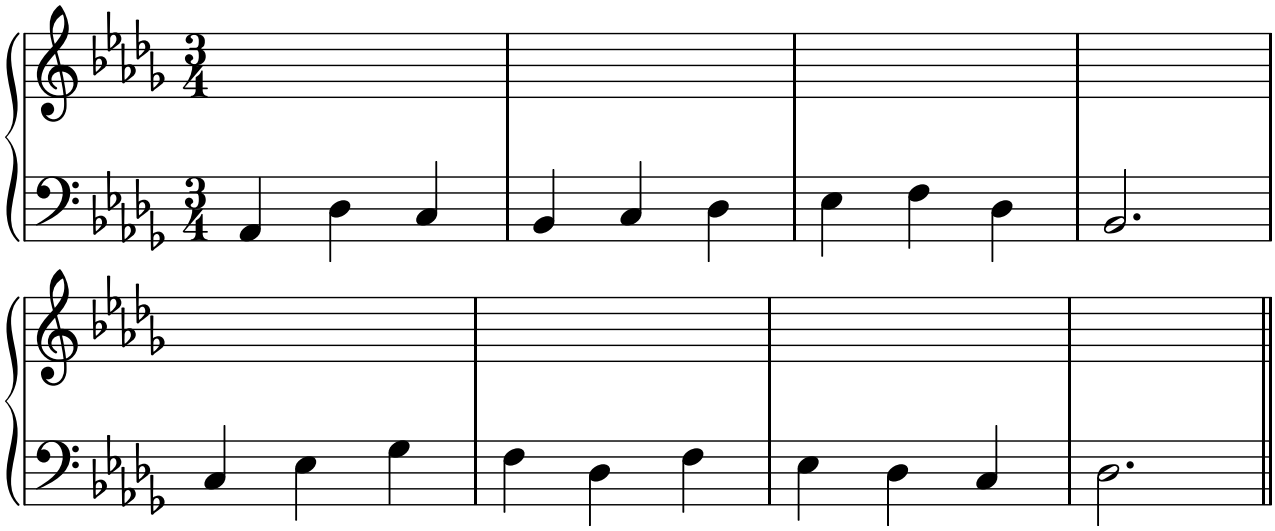


Exercise 2a. 2:1 counterpoint
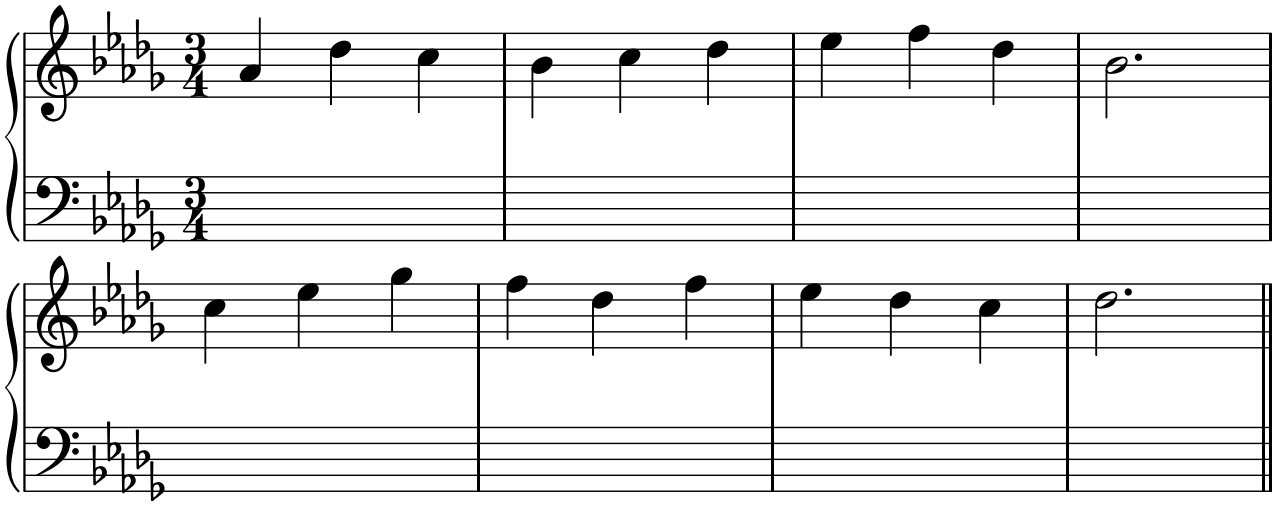


Exercise 2b. 2:1 counterpoint



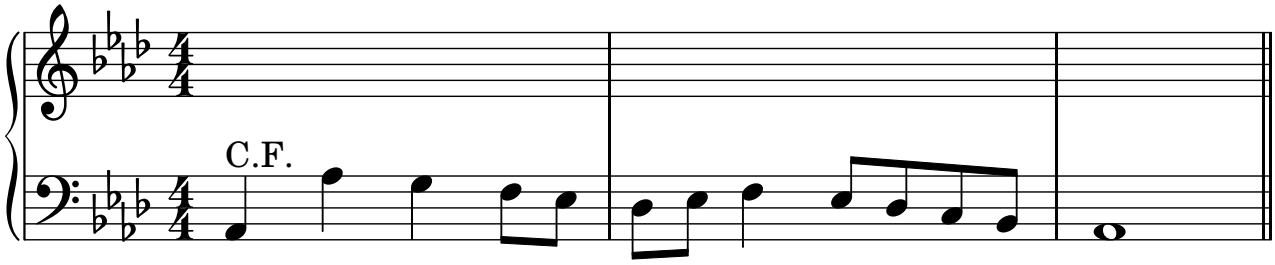
Exercise 3a. 2:1 counterpoint



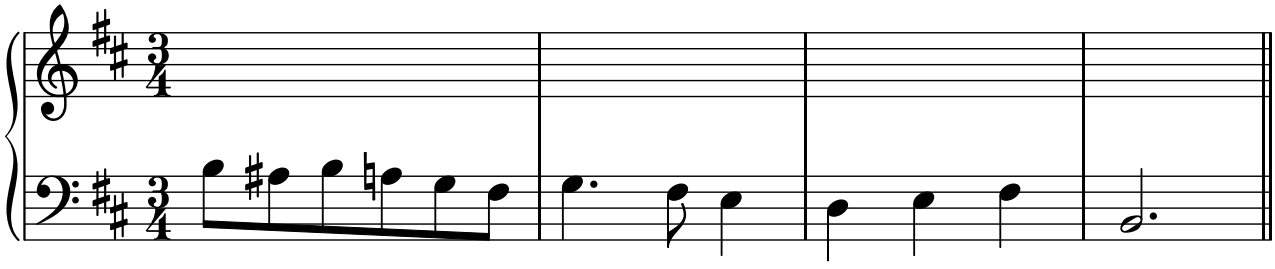
Exercise 3b. 2:1 counterpoint



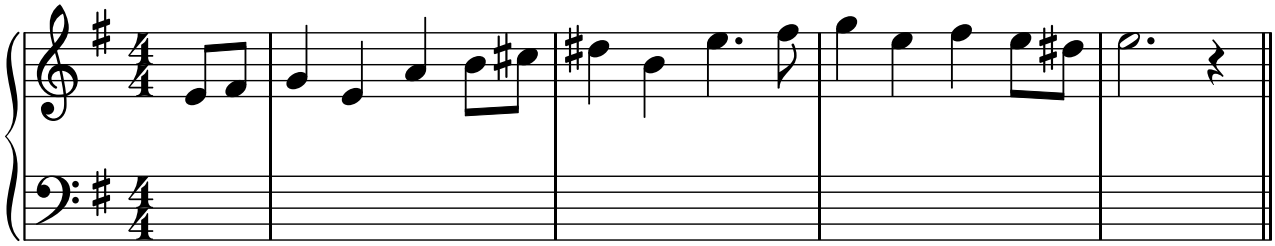
Exercise 4: When there are 8th notes in the cantus firmus, write quarter or 8th notes in the added voice.



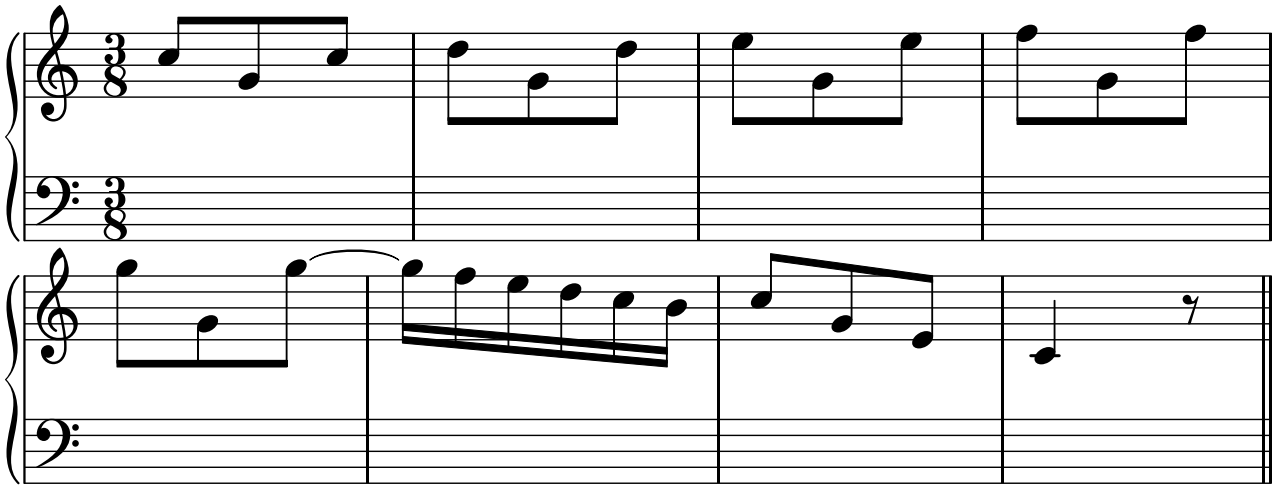
Exercise 5: When there are 8th notes in the cantus firmus, write either quarter or 8th notes in the added voice.
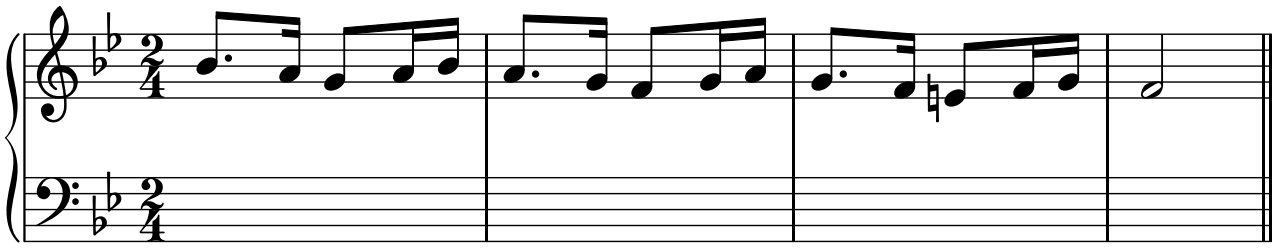


Exercise 6: When there are 8th notes in the cantus firmus, write either quarter or 8th notes in the added voice.



Exercise 7: When there are 16th notes in the cantus firmus, write 8th notes in the added voice.



Exercise 8: Use any combination of 2:1 or 1:1 counterpoint.



Chapter 5

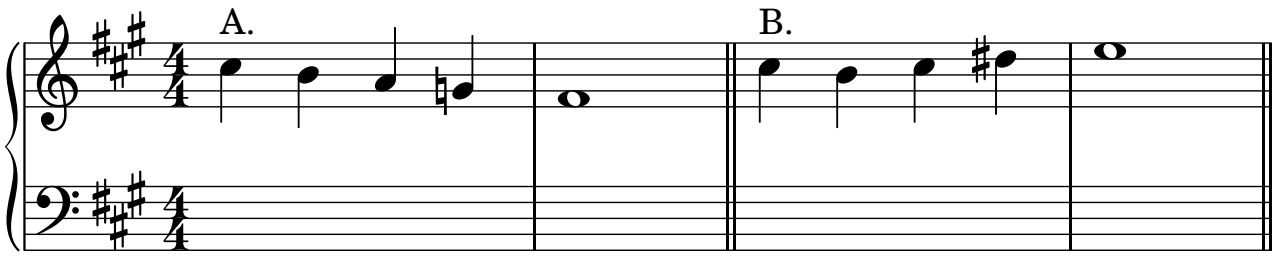
Chromaticism in Two Voices

Chromaticism is used either to imply harmonies outside the key or for melodic embellishment. Here are suggestions for how to deal with chromatic pitches when you find them in a cantus firmus, or how to use them when writing your own counterpoint.

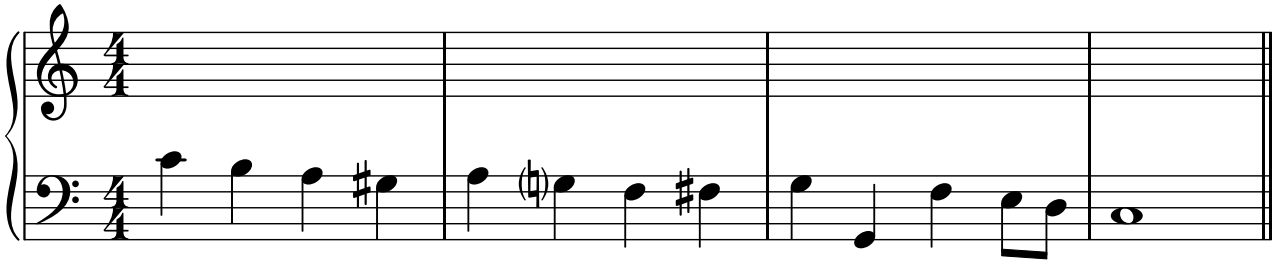
- Treat a raised pitch as the **third** of a secondary dominant—supply either the root or the 7th in your added voice.
  - The 7th creates a tritone and makes for strong harmonic movement.
- Treat a lowered pitch as the $\mathbf{7^{th}}$ of a secondary dominant—supply third in your added voice.
- Be sure to make tendency tones follow their tendencies.
  - Raised pitches tend to move upward by step.
  - Lowered pitches tend to go down by step.
  - Leading tones (even temporary ones) must go up by step.
  - The 7th of a chord resolves downward by step.
- Play through the cantus firmus three or four times to get a sense of its harmonic implications before trying to write counterpoint against it.

Exercises with Chromaticism

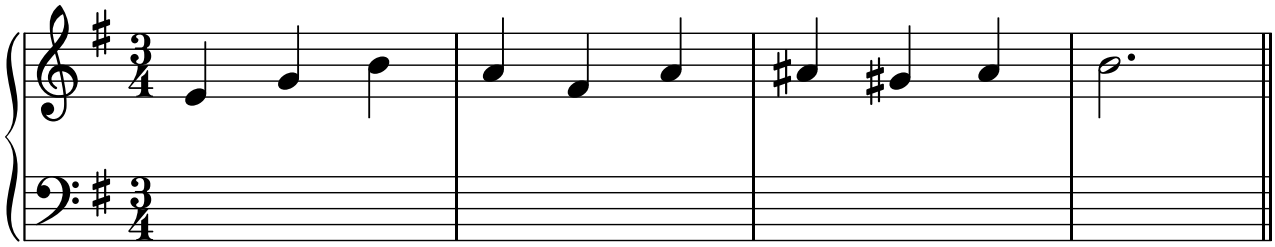
Exercise 1a-1b
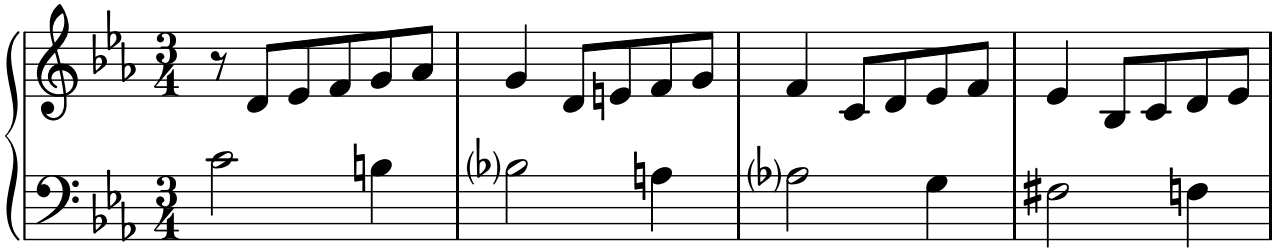


Exercise 2

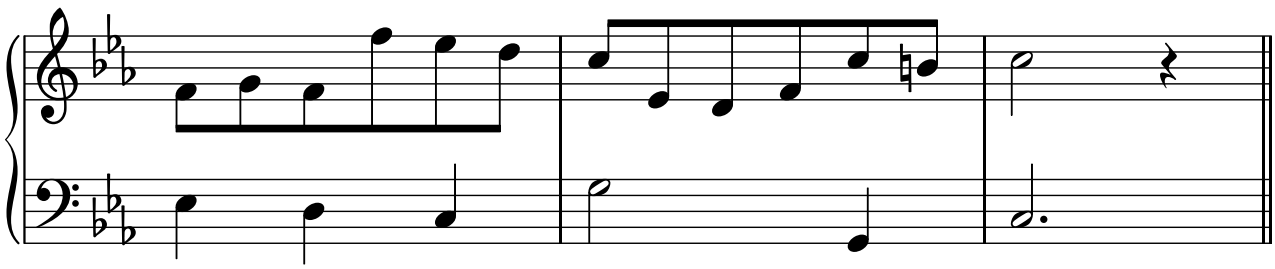


Exercise 3

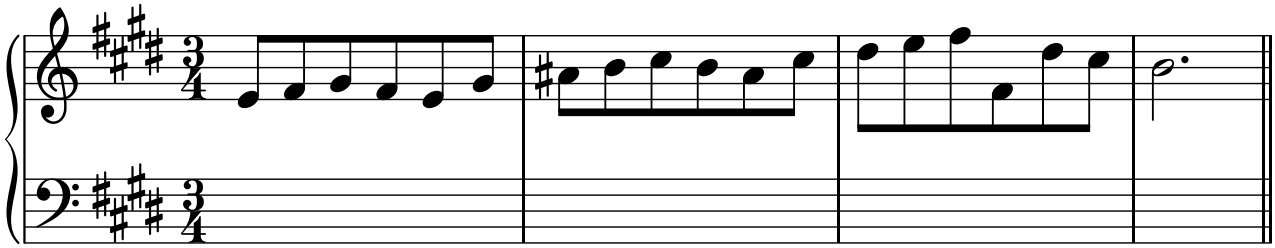


Exercise 4: Added voice should have mostly 8ths and some quarter notes.

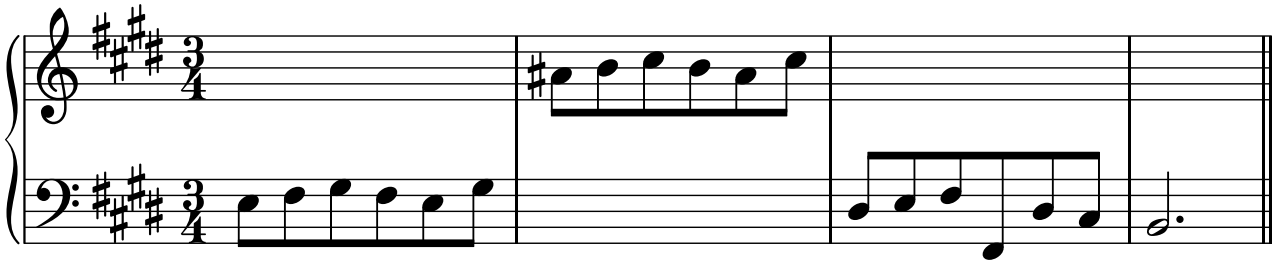




Exercise 5: Any ratio is fine. Where there's a sequence in the given voice, use one in the added voice as well. Note that this example modulates.



Exercise 6: Any ratio is fine.



Chapter 6
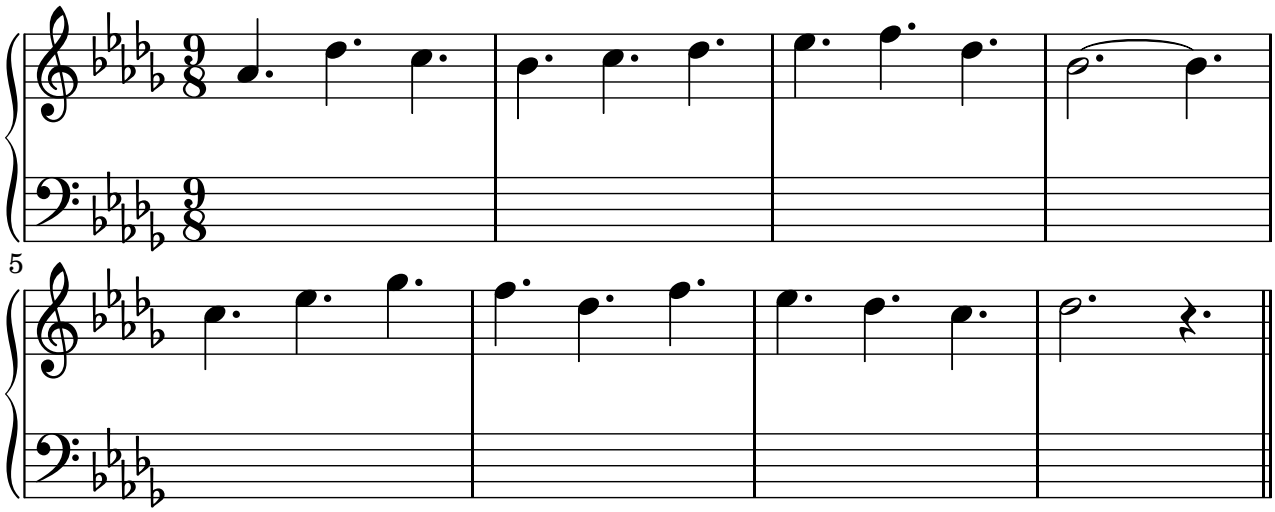
3:1 and 4:1 Counterpoint

Third-species (3:1) counterpont is most often found in pieces having compound meters like $\frac{6}{8}$ or $\frac{9}{8}$. Just as in 2:1 counterpoint, you are strongly encouraged to use dissonance (correctly!).

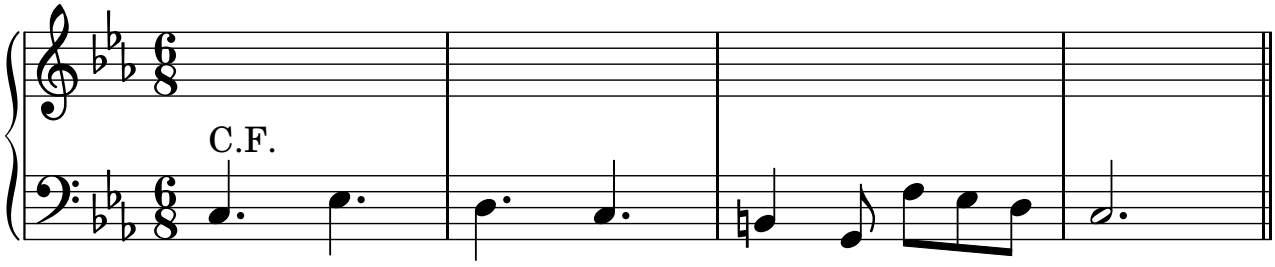
Tips & Warnings

- Avoid stepwise motion followed by a leap in the same direction.
- Watch out for sneaky parallels.
- Don't revolve around the same pitches over and over.
- Don't use too many different patterns in the same exercise.
- No repeated tones—these change the relationship from 3:1 to 2:1

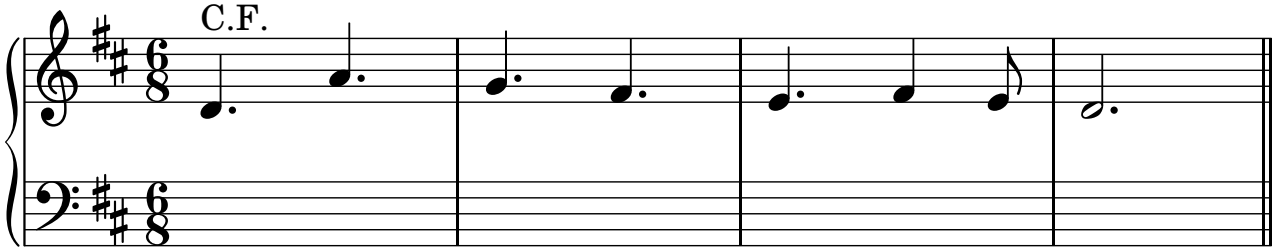
Exercise 1: Write 8th notes against the cantus firmus.



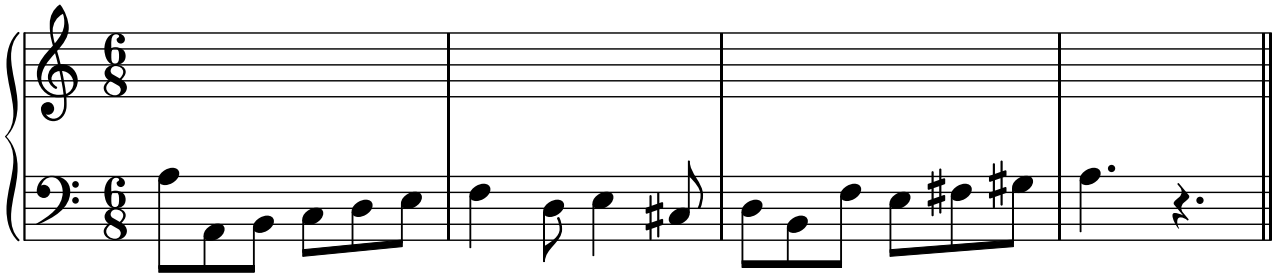
Exercise 2: Write 8th notes against the cantus firmus.



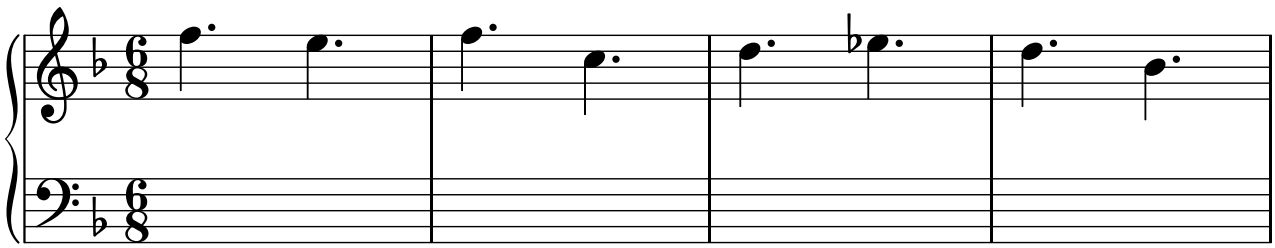
Exercise 3: Write 8th notes against the cantus firmus.

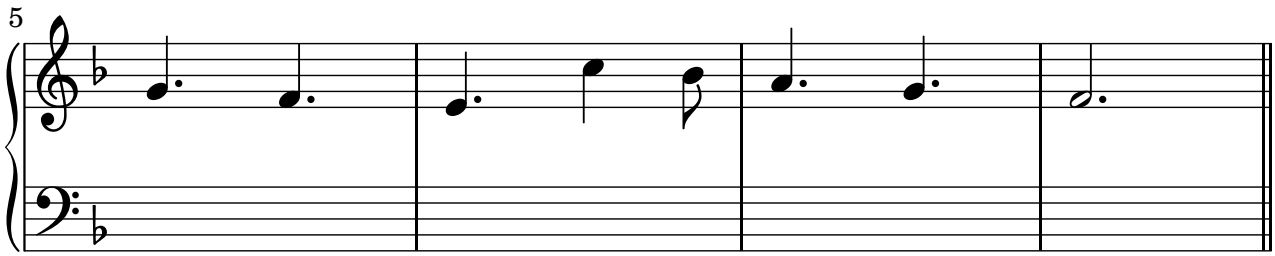


Exercise 4: Use any combination of 3:1 or 1:1 counterpoint.

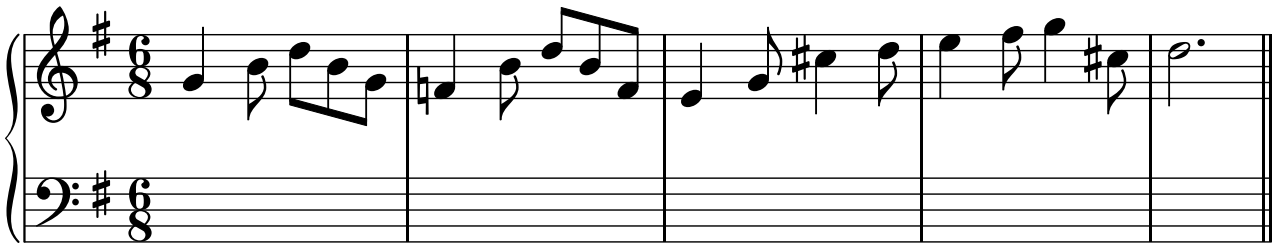


Exercise 5: 3:1 counterpoint





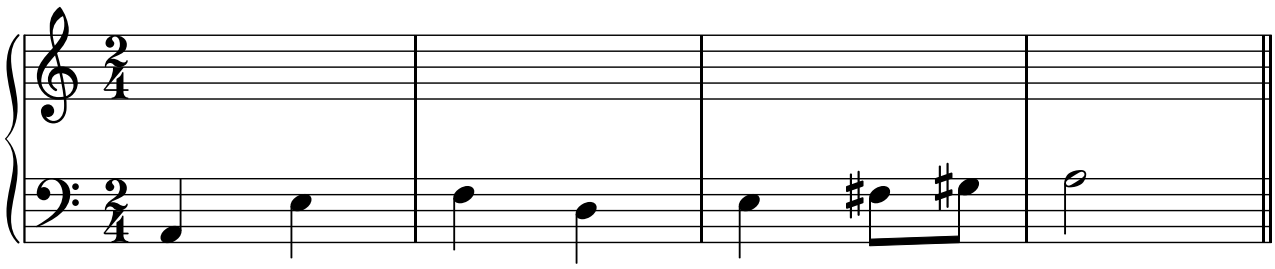
Exercise 6: Use any ratio of counterpoint.
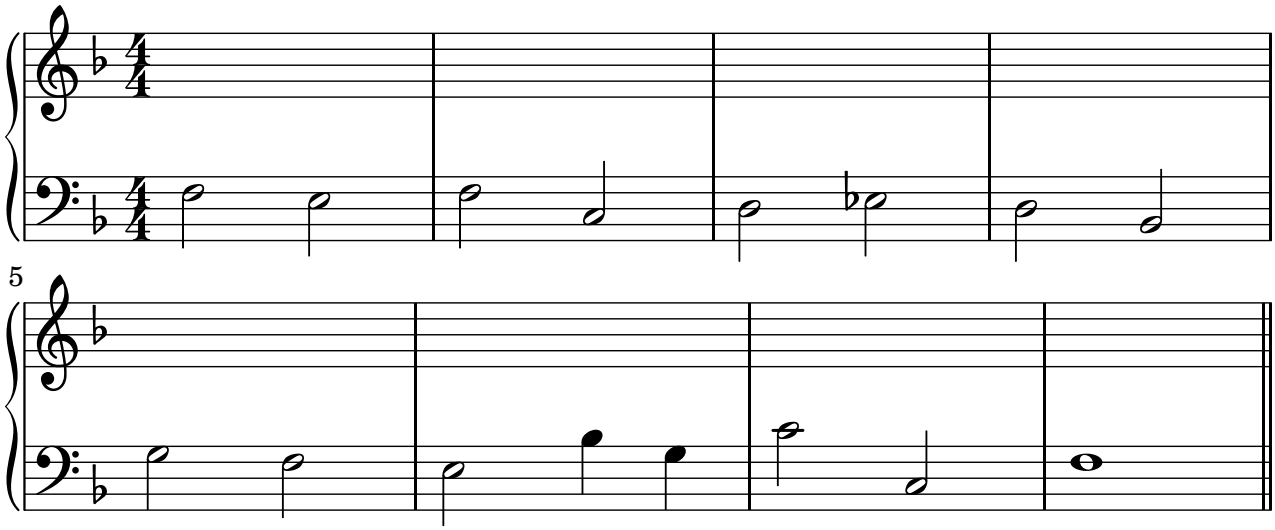


4:1 Counterpoint

The same principles apply in 4:1 counterpoint as in 3:1 and 2:1.

Exercise 1: Write 16th notes against the cantus firmus



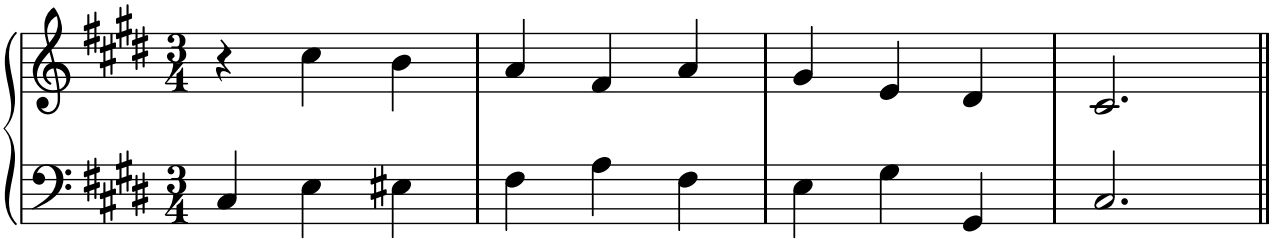
Exercise 2: Write 8th notes against the cantus firmus



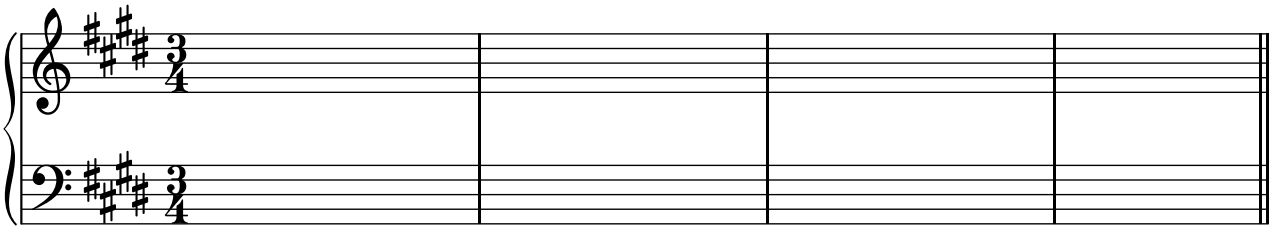
Conversion of 1:1 to other species

Convert the following 1:1 counterpoint to other species. Spread the rhythmic activity between the voices and keep a steady stream of the smaller value of notes.

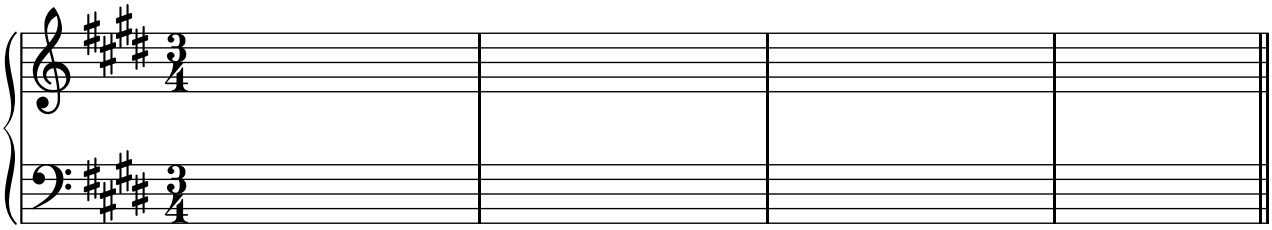
Basic 1:1 Counterpoint



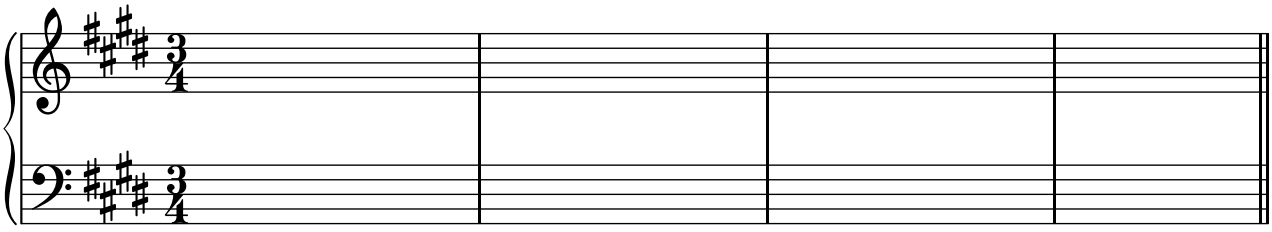
Convert to 2:1 Counterpoint



Convert to 3:1 Counterpoint



Convert to 4:1 Counterpoint

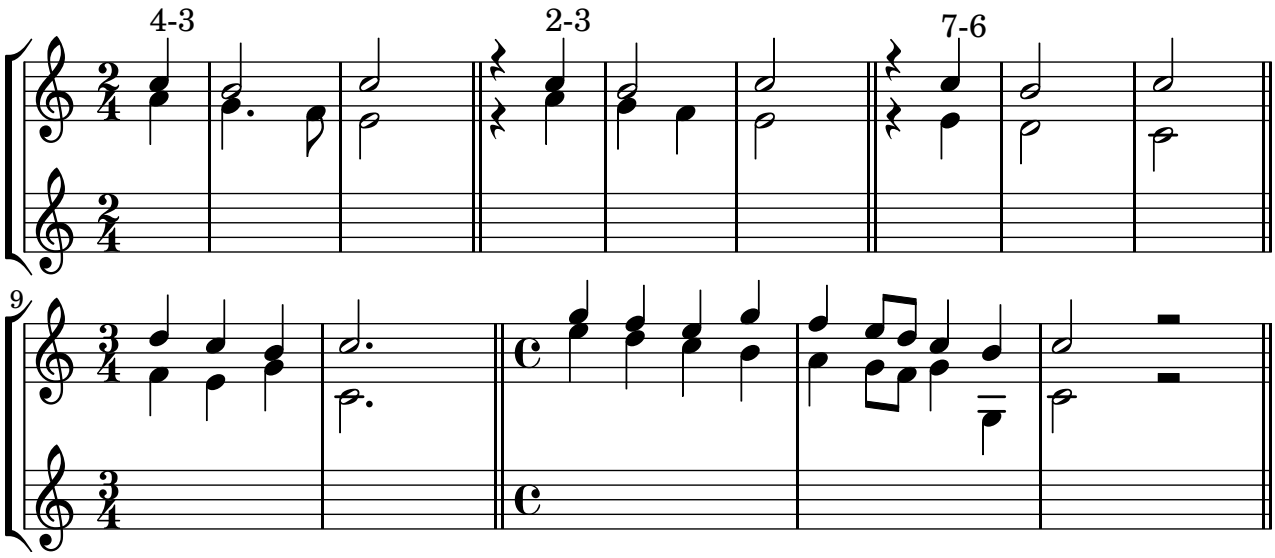


Suspensions

One of the nicest ways to make counterpoint more interesting is to use suspensions. Any time there are parallel thirds or sixths, it is not difficult to introduce 4–3 or 7–6 suspensions, respectively. It is also possible to introduce 2–3 suspensions (where the suspended voice is in the bass) when there are parallel thirds.

Change the following brief passages to include suspensions where possible.

Make Suspensions

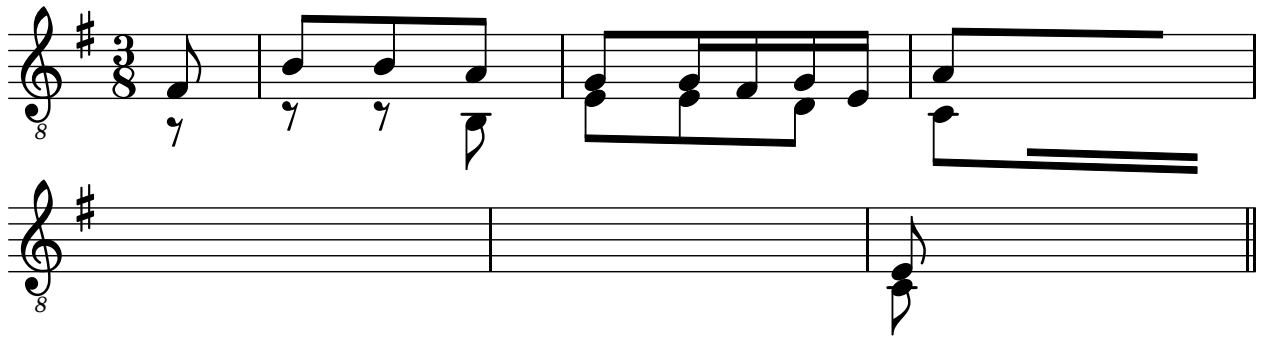


Sequences

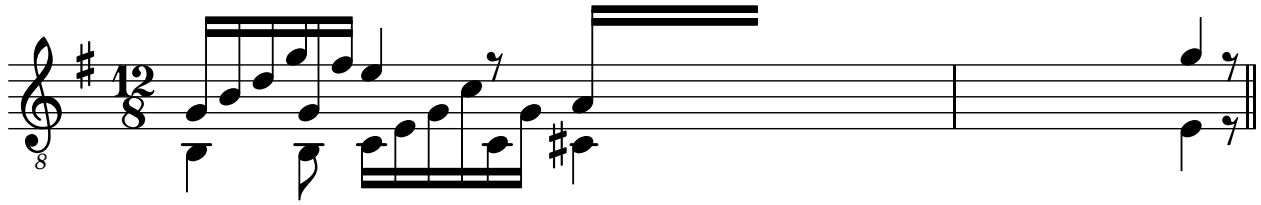
The Baroque aesthetic is based on *unity of affect*, such that a single musical idea is sustained for the duration of a piece. You'll find as a consequence that Baroque composers make frequent use of **sequential repetition** to sustain and string out musical ideas and especially to help effect modulations.

Complete the sequences in the following passages.

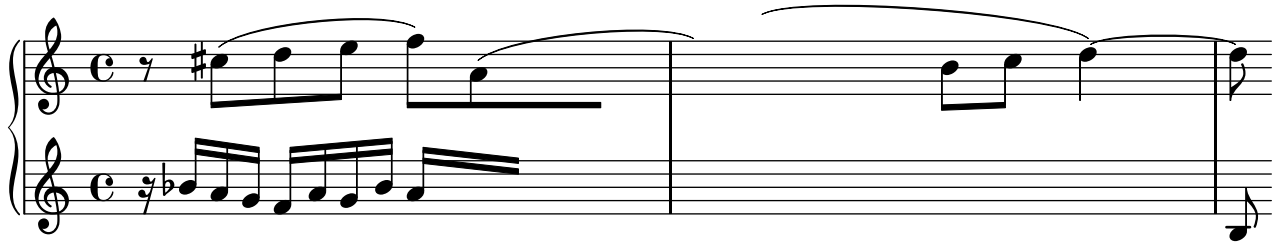
Exercise 1
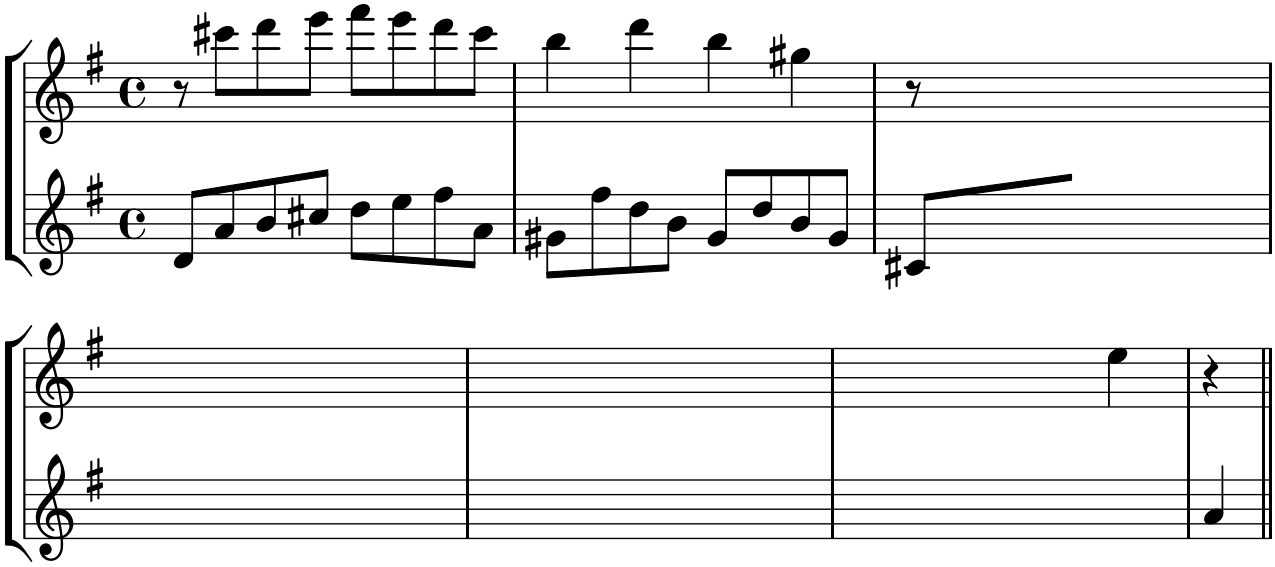


Exercise 2



Exercise 3



Exercise 4
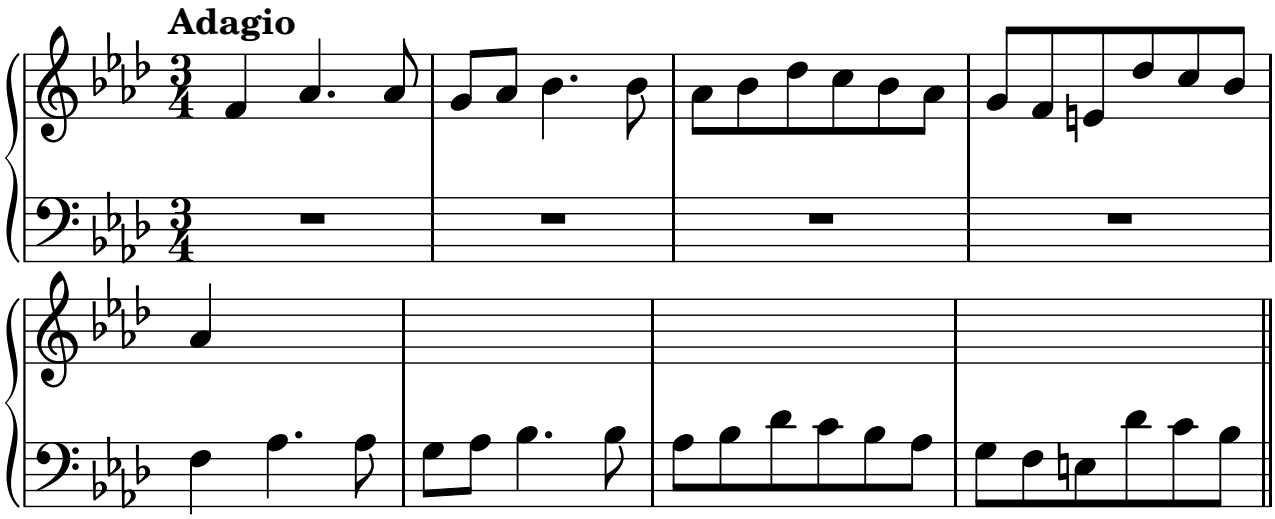


Chapter 7

Invertible Counterpoint and the Invention
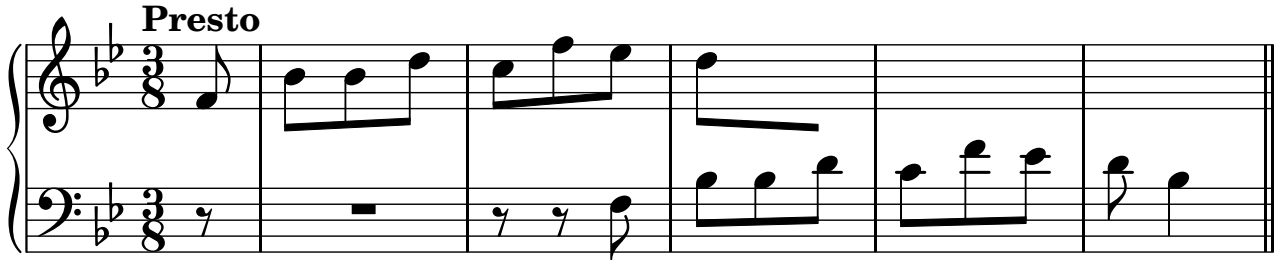
Invertible counterpoint (a.k.a. "Double Counterpoint") is counterpoint that works well no matter which way the voices are situated—the top voice could go on the bottom, the bottom could go on top, and it would sound good either way. It is important to use imperfect intervals when writing invertible counterpoint, as thirds become sixths and vice-versa. Likewise it is important to avoid perfect intervals. Fourths become fifths, and fifths become fourths. In your project on writing Two-Part Inventions, you will have to use invertible counterpoint between your motive and countermotive. Here are some exercises for writing such countermotives in invertible counterpoint.

Write a counter-motive for each of the following motives.

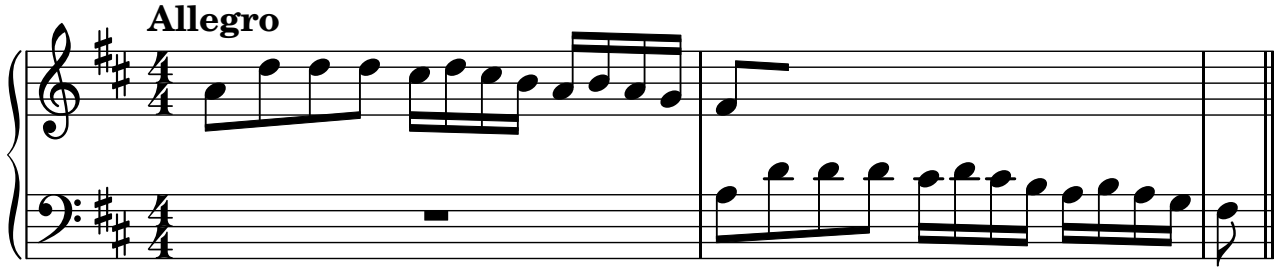
Motive 1



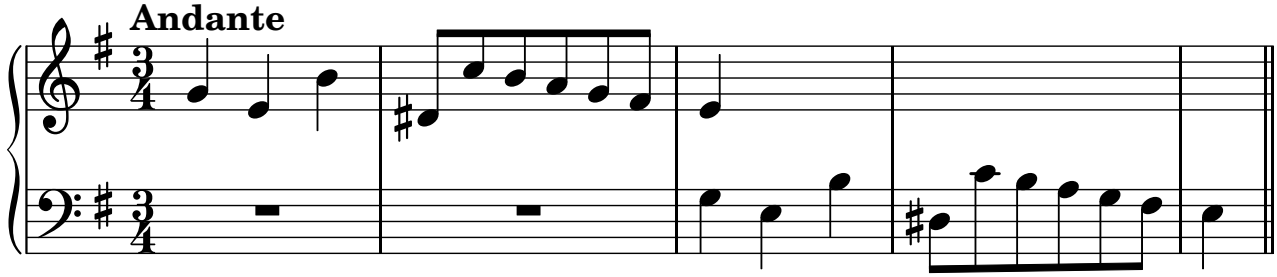
Motive 2



Motive 3



Motive 4



Chapter 8

Counterpoint Projects

One of the best ways to improve contrapuntal skills is to try your hand at writing a real piece in 18th-century style. You will enjoy the freedom of being able to mix the species of counterpoint you have practiced in the exercises, and you will begin to appreciate the difficulty of writing good melodies, cadences and modulations. In the counterpoint class I took at the University of Texas we wrote five pieces meant to approximate styles and genres of the late baroque era:

1. Organ Variations
2. Binary Dance Piece
3. Two-Part Invention
4. Three-voice Fugue
5. Choral Fugue Exposition

The guidelines for each project are taken almost verbatim from those given to us by Dr. Patrick McCreless, our instructor for the class.

A Note about Cadences

Many students have trouble writing effective cadences. Good cadences have the following characteristics:

1. A strong harmonic progression
   - `ii-V-I`
   - `IV-V-I`
2. Clear definition by the bass line
3. Strong metric placement

Tips for Writing Cadential Trills

- Trills start on upper note
- A cadential trill typically ornaments a `V` harmony (sometimes with $\frac{6}{4}$ harmonic ornamentation)
- Trill happens on either the supertonic or leading tone in upper voice at a `V-I` cadence
  - Leading tone (3rd of a dominant harmony) is best in 2 voices
  - Supertonic (5th of a dominant harmony) sounds o.k. too
- Trilled note often leads to an **anticipation** (type of non-chord tone).

Here are examples of cadential trills on both the leading tone and the supertonic. In each case you can see how it is typically written in the score (with the "Tr " indication) and how it might be realized by the performer. Notice the anticipation of the tonic in each case.

Figure 8.1: Cadential trill on the leading tone

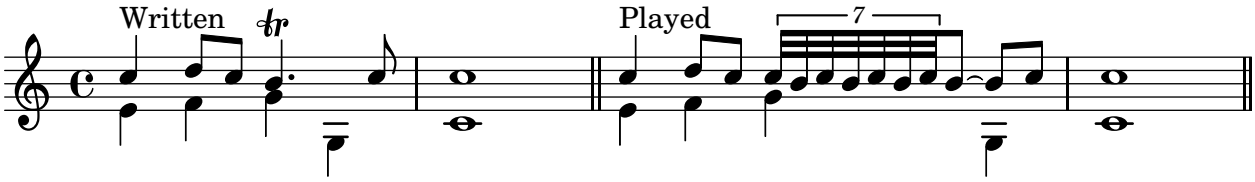
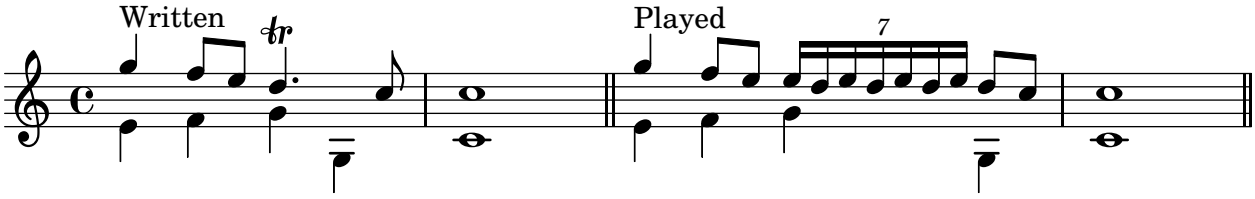


Figure 8.2: Cadential trill on the supertonic



Project 1: Binary Dance Piece

Write a two-voice binary piece of at least twenty-four measures. Some specifics:

- The piece must be in a key with two or more sharps or flats in the key signature; it may be in either major or minor mode.
- It must be a continuous binary form: that is, if in major mode, it must modulate to `V` at the cadence before the double bar; or if in minor mode, it must modulate to `III`.
- It must be in **rounded binary** form.
- In the section after the double bar, it **must modulate** to at least one other closely-related key.
- The phrasing should be generally regular–four-measure phrases should be the norm, with occasional two- or six-measure phrases if you like.
- Adjust the difficulty of the piece to your compositional experience and ambition. If you have written pieces like this before, try a Courante or Bourree in the style of the Bach keyboard suites. If you have not, and if you find the task difficult, write a Minuet or other relatively straightforward dance piece.

Possible Dance Types for your project

- Allemande
- Courante
- Sarabande
- Minuet
- Gavotte
- Gigue
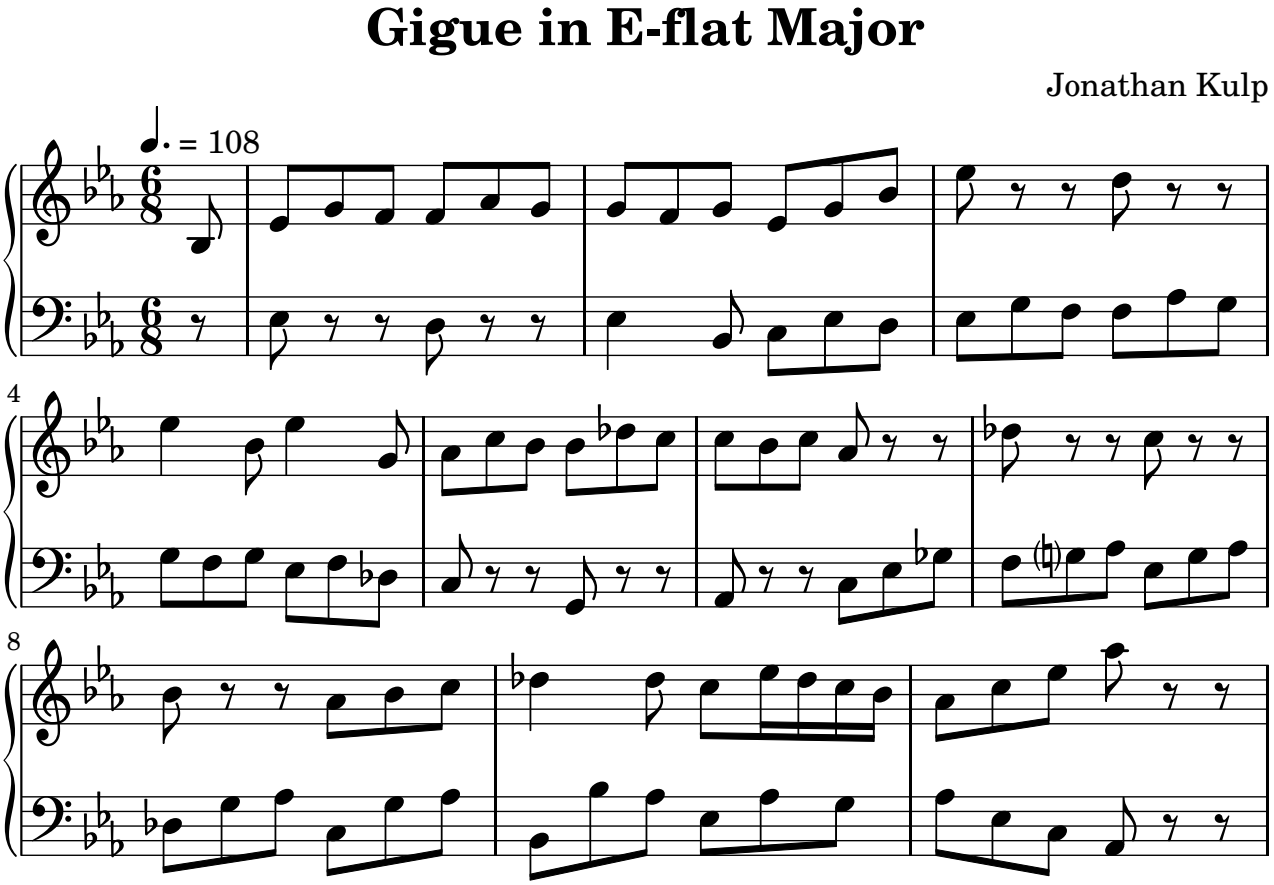
For characteristics of the various dance types, check *The New Harvard Dictionary of Music* or *The New Grove Dictionary of Music and Musicians*, and perhaps look at some of the suites of J.S. Bach. For whatever dance type you choose, it needs to have the right characteristic meter, tempo, and rhythmic gestures. (If you call your piece a gigue, it needs to sound like a gigue!)

Tip!

If you get into a jam, try working backwards. For example, if you know that you must have a cadence that lands on `V` on the downbeat of m. 8, then write the `V` on the downbeat of m. 8. Next, sketch out a good harmonic progression for the bar preceding it and fill in the notes that will make a convincing cadence. Working backwards like this you can quickly hook up with the first few bars you wrote.

What's Wrong with This?

A couple of years ago I decided that I would share my students' pain and write a binary piece along with them when they were working on their projects. Since I'm more experienced than they are, I thought I'd write a gigue, and it was going great until I got about 10 bars into the piece and I gave up. See if you can see **why** I abandoned it, and better yet, how can this piece be saved? Good luck!



Project 2: Two-Part Invention

Write a two-part invention in the style of Bach. Guidelines:

1. Use a short motive and countermotive in imitation at the beginning, and use the two at least twice in **invertible counterpoint** later in the piece.
2. The piece should be at least sixteen measures long, with primary movement in sixteenth notes.
3. The invention must modulate to `V` and articulate that motion with a standard cadence within the first five or six measures. (Minor keys: to `III` or `v`)
4. The invention must **modulate to two other closely-related keys** after the modulation to `V` (or `III`).
5. The invention must include at least one **sequence** in the middle section.

Keys to Success

Use Bach's Two-Part Inventions as models. Try to come up with a motive that is simple, but has a distinctive character. Think in terms of long-range linear and tonal goals. Before getting too deep into the piece, be sure that your motive and countermotive work perfectly in invertible counterpoint. This will save you work later. Play the piece back often as you go along so that you have a sense of how it's progressing.

Project 3: Fugue

Write a three-voice fugue of twenty or more measures. Guidelines:

1. It must have a subject and two countersubjects, the three of which must work in triple counterpoint. You must use this invertible counterpoint at least twice in the fugue, in addition to its initial appearance.

2. You should have a recognizable, standard Baroque-style cadence in V (major keys) or III (minor keys).

3. There must be a modulation to at least one closely-related key other than V (major keys) or III (minor keys) in the middle section of the fugue. One such key must be articulated by a cadence. You will be penalized for modulations to keys other than closely-related keys.

4. The fugue must end with a conclusive, root position, V-I, standard, unmistakable cadence. No wimpy final cadences![1]

Suggestions

- Do some analysis of Fugues from the *Well-Tempered Clavier*. The fugue in C minor from Book I is a great example.

- For episodes, take a fragment of your subject and use it in sequential repetition to modulate to new keys.

- Chapters 15-16 of the Kennan book should help answer questions about Fugue and fugal procedures.

[1] This was one of Dr. McCreless' all-time great directives, given out of frustration with our lame cadences.

Project 4: Choral Fugue Exposition

Write the exposition of a four-voice choral fugue using the following text:

```
"Benedictus qui venit in nomine Domini"
```

Guidelines

1. The exposition should be in $\frac{4}{4}$ time, with a subject of four measures. The subject should begin somewhat slowly, but eventually move into 16th notes.

2. You should be sensitive to the fact that you are writing vocal music: the parts should be in a singable vocal range, and should not be too disjunct, especially in sixteenth-note passages.

3. There must be at least one countersubject. That is, each voice, after it states the subject, must proceed to the countersubject. You do not have to have a second countersubject.

4. The voices may enter in any order

5. Once all four voices are in, proceed to a cadence in either the tonic or the dominant (or III in minor keys). Also, once the four voices are in, do not drop any out.

6. The exposition should be about twenty measures long.

Project 5: Fugue Analysis

The final project is an analysis of Bach fugue of your choice or one of the two fugues suggested below from Book II of the WTC. I will provide you with an electronic copy of the suggested pieces (.pdf format), as well as a recording that's available on the shared listening list.

What to find and mark in the score

- All appearances of the subject
- All appearances of the countersubject (if any)
- All appearances of the second and third countersubjects (if any)
  - In the exposition, label subjects and answers
  - Indicate whether it's a tonal or real answer
- Transformations of the subject
  - retrogrades, inversions, and augmentations
  - other tricks like stretto and stuff like that
- Mark off the episodes with brackets
- Indicate where the melodic material of the episodes comes from, if it's clear at all
- Indicate key areas for each appearance of the subject

Suggested Fugues

1. WTC Book II, Fugue no. 5 in D major
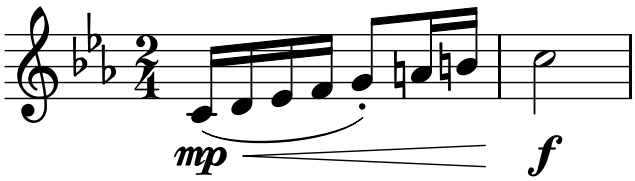
Colophon

The main text of this book was typeset using the LaTeX typesetting engine. It is a free program used by publishers in the fields of science and mathematics because of its excellent output of formulae. I decided to use LaTeX for this project not only because of its beautiful output, but also because it supports musical accidental symbols perfectly (♭,♯,♮) without messing up line spacing. LaTeX also makes it easy to produce scale-degree numbers with the little carets on top of them. For example $\hat{5}$ is produced simply by typing `\^5`. It's also easy to create inline time signatures, such as $\frac{12}{8}$ and $\frac{3}{4}$.

LaTeX integrates well with **Lilypond**, the free/open-source music-typesetting program I used to create all of the musical examples and exercises in the book. Notes and all other symbols are entered in plain text files, which are then compiled by the typesetting engine and turned into images. Notes are entered by their names (`a b c`), and then durations (`2` for half note, `8` for 8th, etc.), dynamics (`\mp` and `\f` in this example), articulations and other objects are attached to the notes. The following simple snippet of Lilypond code shows these features.

```
\version "2.17.97"

\relative c' {
  \key c \minor
  \time 2/4
  c16(\mp\< d es f g8)-. a16 b
  c2\f
}
```



For information about Lilypond visit `http://lilypond.org`. Lilypond runs on Linux, Windows, and Mac platforms and is available for free download at the website.

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